

# **Project Report:**

# **MediCare**

# **Connect**

# **(Technova)**

## 1. Executive Summary

**MediCare Connect** (hosted in the [Technova](#) repository) is a modern, full-stack healthcare management platform designed to digitize administrative and clinical workflows. Built using modern web technologies like **TypeScript**, **Supabase**, and a dual **Frontend/Backend** architecture, the system establishes a secure, role-based ecosystem connecting patients, doctors, medical staff, and administrators.

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## 2. Project Overview

- **Repository:** [praveenkumars-cmd/Technova](#)
  - **Primary Language:** TypeScript (93.1%), JavaScript (6.1%)
  - **Core Architecture:** Full-stack architecture divided into frontend/ and backend/ micro-services/modules.
  - **Database & BaaS:** [Supabase](#) (Authentication, Database, Real-time updates, and Storage).
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## 3. Key Features & Functionality

### Role-Based Access Control (RBAC)

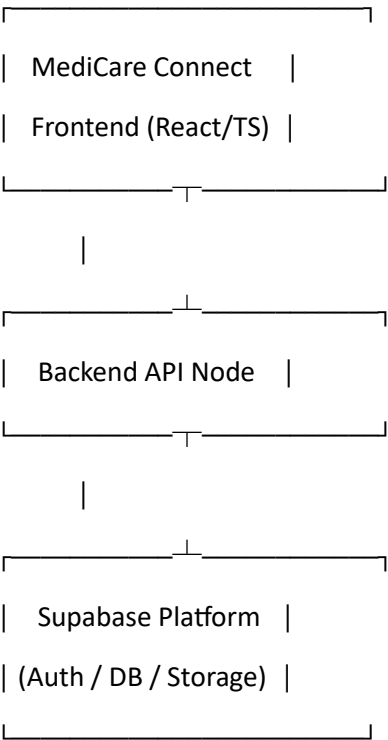
Dedicated portals and capabilities tailored for four core user roles:

- **Patients:** Schedule appointments, view electronic health records (EHR), access prescriptions, and manage personal profiles.
- **Doctors:** Manage patient appointments, issue digital prescriptions, review medical history, and log session notes.
- **Staff:** Handle front-desk check-ins, manage appointment queues, and update patient registration details.
- **Admins:** Oversee user permissions, platform analytics, system logs, and security governance.

### Healthcare Workflows

- **Appointment Management:** Real-time scheduling, status updates, and calendar coordination.
  - **Electronic Health Records (EHR):** Digitized, secure patient medical history and diagnostic records.
  - **Digital Prescriptions:** Doctor-generated prescriptions tied directly to patient profiles.
  - **Analytics & Notifications:** Insights into clinic operational efficiency and automated alerts for upcoming appointments or medical updates.
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4. Technical Architecture & Stack



| Component       | Technology / Stack             | Role                                                                                                |
|-----------------|--------------------------------|-----------------------------------------------------------------------------------------------------|
| Frontend        | React / Next.js / TypeScript   | User interface, role-specific client dashboards, and responsive forms.                              |
| Backend         | Node.js / Express / TypeScript | API endpoints, business logic execution, and server-side validation.                                |
| Database & Auth | Supabase (PostgreSQL)          | Managed database, authentication middleware, and secure data isolation.                             |
| Documentation   | SRS Document                   | Included software requirements specifications ( <a href="#">SRS SOFTWARE HACKATHON final.docx</a> ) |

## 5. Repository Structure

Plaintext

Technova/

|                                      |                                                      |
|--------------------------------------|------------------------------------------------------|
| └─ backend/                          | # API routing, business logic, Supabase integrations |
| └─ frontend/                         | # Client-side UI components and user dashboards      |
| └─ SRS SOFTWARE HACKATHON final.docx | # System Requirement Specification document          |
| └─ package.json                      | # Project dependencies and setup scripts             |
| └─ tsconfig.json                     | # TypeScript compiler configurations                 |
| └─ .gitignore                        | # Git tracking exclusions                            |
| └─ README.md                         | # Overview and setup guidelines                      |

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## 6. Recent Developments & Milestones

- **Database & Auth Integration:** Integrated Supabase backend with frontend services for unified authentication and database querying.
- **Software Requirements Specification:** Added formal SRS hackathon documentation to outline workflow parameters and system architecture.